

CSI5324: Software Engineering

General Information

This course provides a comprehensive introduction to the principles, processes, methods, and practices of modern software engineering. It examines software development across the software development life cycle, including requirements engineering, system and software design, implementation, testing, deployment, maintenance, and evolution. The course considers both foundational software engineering concepts and contemporary approaches to developing software products in collaborative and iterative environments.

Students will examine how software engineering principles are applied to the development of reliable, maintainable, and high quality software systems. Topics include software processes and agile development, requirements specification, architectural and detailed design, software testing and quality assurance, configuration and version control, project planning, software evolution, and professional software engineering practices. Through team based project work, students will gain practical experience applying software engineering methods and tools to the development of a software product.

Objectives: At the end of the course, students should be able to

- **Explain** fundamental software engineering principles, processes, and practices across the software development life cycle.
- **Analyze** software requirements and apply appropriate methods for software design and development.
- **Apply** software testing, quality assurance, and maintenance practices to improve software reliability and maintainability.
- **Use** modern software engineering tools and collaborative development practices in a team based environment.
- **Evaluate** software engineering decisions and tradeoffs related to quality, cost, schedule, maintainability, and professional responsibilities.

When and Where:

- Lectures: Tuesday & Thursday 11:00 AM – 12:15 PM @ Cashion C315

Instructor: Dr. Mengfei Ren

- Office Hours: TR 10 -11 am or by appointment.
- Office: Cashion 304
- E-mail: [Mengfei_Ren\(AT\)baylor.edu](mailto:Mengfei_Ren(AT)baylor.edu)

Recommended Textbook

No required textbook in this course, but we mainly cover the topics from the following two textbooks and other materials:

- **Engineering Software Products:** An introduction to modern software engineering– Ian Sommerville

- **Software Engineering** (10 edition): A comprehensive textbook on software engineering – Ian Sommerville

Course Topics

Schedule: The class schedule will be posted on the Canvas Syllabus page. The schedule is updated on a regular basis, so use the "current" schedule information (rather than a printed copy) to determine what the class is doing.

Course Requirements

- **Design Studio:** There will be several in-class or take-home design studio assignments through the semester, involving written/modeling work and programming. Tentative deadlines for the assignments are being worked out and will be available soon. (Please regularly check out the Assignments page on Canvas.) You lose 25% of the grade for each extra calendar day of late submission. After 4 days, you won't get any credit for the assignment unless you obtain prior permission from the instructor. **However, watch out that some assignments will be given without any late submissions allowed.** So, it is **your responsibility** to check the submission details when each assignment is posted.
- **Software Development Tool Presentation:** Each student will select a software development application, tool, or technology and give an individual presentation and demonstration to the class. The presentation should introduce its purpose, key features, and role in the software development process. Students will demonstrate how the selected application can be used in a practical software development context.
- **Team Project:** Students will work in teams to develop a software application over the course of the semester. The project will provide hands on experience with major software development life cycle activities, including requirements analysis, design, implementation, testing, and deployment. Teams will use appropriate software engineering tools and practices for collaboration, version control, documentation, and project management. Project evaluation will consider both team outcomes and individual contributions.
- **Quizzes:** In-class quizzes are scheduled ones (i.e., each quiz has its due date). Pop quizzes are not announced, though. Both will cover the assigned reading or topics already covered in class. (NOTE: The instructor may ask questions about readings that have not been specifically discussed in class.)
- **Reading Assignments / Reviews:** Students are required to complete reading the listed chapters by the end of the week indicated. Not all the materials will be covered in depth in the class, but students will be tested on the contents of these chapters. Review problems will be given to help and test several selected chapters of reading assignments. Refer to the Course Schedule page posted on Canvas for more detail.

Grading Information:

Marks will be allocated as follows:

- 60% - Team Project
- 10% - Design Studios
- 5% - Tool Presentation
- 5% - Quizzes
- 20% - Final Exam (cumulative), Friday Dec 11, 2-3:15pm

A: 90-100, B: 89-80, C: 79~70, D: 60~, F: below 60

Other Policies

Late Submission & Make-ups

- Late assignments lose 25% per day. After 4 days, you will not receive any credit without prior instructor permission.
- If you miss an exam or a presentation due to the illness or personal emergency, a written explanation (including supporting documentation) must be submitted to the instructor. If the explanation is acceptable, an alternative to the activity will be arranged. Make-up arrangements must be arranged prior to the scheduled due date.

Course Communication:

- **Canvas:** All assignments, lecture notes, and quizzes will be posted on Canvas, so check the online schedule on Canvas regularly (at least twice a day).
- **Email:** Although we will use Canvas for most correspondence, some important information may be sent via email. So, please **read your Baylor email regularly** although I will try to keep emails to a minimum.

Your Responsibility: You are responsible for:

- knowing what goes on during class; if you miss a class, talk to someone who did attend or ask the instructor
- "tracking down" a handout if you miss class; old handouts will not be brought to class
- knowing the class schedule -- in particular, what assignments are due; the class schedule is posted on a web page (see above) and is updated on a regular basis -- so, use the "current web page" (rather than a printed copy) to determine what the class is doing
- reading what is associated with a particular topic; explicit reading assignments will not be given in class (see the class schedule to determine the topic of the day and its associated reading); **NOTE:** this is important with respect to quizzes.

Artificial Intelligence (AI)

- Unless I instruct you otherwise, all assignments should be your original work and should not be produced in part or in total with the assistance of artificial intelligence (for example, ChatGPT,

Grammarly, or some other resource). Use of artificial intelligence without explicit permission from the instructor constitutes a violation of the Honor Code at Baylor University. I may at times use artificial intelligence tools for instruction or allow or expect you to use them to complete your assignments, but in these instances, I will provide clear permission and guidance for their proper use.

Attendance:

- If you are tardy to class, it will only count as 75% of a presence.
- If a student is unable to attend class because of an illness or some other extreme circumstance, it is her/his responsibility to notify the professor (by e-mail or phone) before class begins. All assignments must be turned in on time. No assignments will be accepted after the due date. There will be no make-up work or extra credit.

Electronic Devices during Lecture:

- **Computers and phones (including texting) are not allowed during lecture time unless you are instructed to use them by the instructor. It distracts you and those around you.**
- If you are seen using your cell phone during class, you will be asked to leave class for that day.
- The instructor will have a cell phone, however, in case of emergencies.
- For selected class activities such as design studios, the use of computers or tablets can be permitted, but you should still remain engaged in class/group activities.

Academic Integrity: Plagiarism or any form of cheating involves a breach of student-teacher trust. This means that any work submitted under your name is expected to be your own, neither composed by anyone else as a whole or in part, nor handed over to another person for complete or partial revision. Be sure to document all ideas that are not your own. Instances of plagiarism or any other act of academic dishonesty will be reported to the Honor Council and may result in failure of the course. Not understanding plagiarism is not an excuse. As a Baylor student, I expect you to be intimately familiar with the Honor Code at: <https://honorcode.web.baylor.edu/>

Baylor University Equity, Civil Rights, and Title IX*Civil Rights Policy and Sexual and Interpersonal Misconduct Policy*

Baylor University does not tolerate unlawful harassment or discrimination on the basis of sex, gender, race, color, disability, national origin, ancestry, age (over 40), citizenship, genetic information or the refusal to submit to a genetic test, past, current, or prospective service in the uniformed services, or any other characteristic protected under applicable federal, Texas, or local law (collectively referred to as Protected Characteristics).

If you or someone you know would like help related to an experience involving:

1. Sexual or gender-based harassment, sexual assault, sexual exploitation, stalking, intimate partner violence, or retaliation for reporting one of these types of prohibited conduct, please

visit baylor.edu/titleix [Links to an external site.](#), or contact us at (254) 710-8454, or [mailto: TitleIX_Coordinator@baylor.edu](mailto:TitleIX_Coordinator@baylor.edu).

2. Harassment (excluding those issues listed in #1) or adverse action based on Protected Characteristics, please visit baylor.edu/civilrights [Links to an external site.](#), or contact us at (254) 710-7100 or Civil_Rights@baylor.edu.

The Office of Equity and Title IX understands the sensitive nature of these situations and can provide information about available on- and off-campus resources, such as counseling and psychological services, medical treatment, academic support, university housing, and other forms of assistance that may be available. Staff members at the office can also explain your rights and procedural options. You will not be required to share your experience. **If you or someone you know feels unsafe or may be in imminent danger, please call the Baylor Police Department (254-710-2222) or Waco Police Department (9-1-1) immediately.**

Except for Confidential Resources, all University Employees are designated Responsible Employees and thereby mandatory reporters of potential sexual and interpersonal misconduct violations. Confidential Resources who do not have to report include those working in the Counseling Center, Health Center and the University Chaplain, Dr. Burt Burleson.

Students Needing Accommodations

Any student who needs academic accommodations related to a documented disability should inform me immediately at the beginning of the semester. You are required to obtain appropriate documentation and information regarding your accommodations from the Office of Access and Learning Accommodation (OALA). Stop by the first floor of Sid Richardson, East Wing in the Paul L. Foster Success Center or call (254) 710-3605 or email OALA@baylor.edu.

Academic Success

I believe every student who has been admitted to Baylor can be successful, and I want to partner with you to help you thrive academically. Be sure to take advantage of the many resources available for academic success, including coming to see me during my office hours. Students who regularly utilize the great resources in the [Paul L. Foster Success Center](#) [Links](#), such as [tutoring](#) [Links](#), [Learning Lab](#) [Links](#), and [Academic Mentoring](#) [Links](#), are among my most successful students. If your academic performance in this class is substandard, I will submit an Academic Progress Report to the Success Center so that the team of coordinated care professionals can ensure that you get the help you need.

University Writing Center

I encourage you to visit the [University Writing Center \(UWC\)](#) this semester and get feedback on your writing for this course. Located in Moody Library 2nd floor West, the UWC offers free assistance to you at any stage of the writing process (brainstorming, researching, outlining, drafting, revising, editing). In their feedback, the consultants focus on higher order concerns, such as content, thesis, evidence, and organization, before grammar or style. The UWC tutors will not proofread, edit, or write your paper for you, but they will equip you with a toolbox of strategies to improve your writing, research, and editing skills. Please take the assignment prompt, your paper/text, and other materials you might need with you to your appointment. Please include my name as the professor, and a report will automatically be sent to

me after your session. Go to the website (www.baylor.edu/uwc) to schedule an appointment, call the UWC at (254)710-4849, or stop by in person.

First Generation College Students

Baylor University defines a first-generation college student as a student whose parents did not complete a four-year college degree. The First in Line program is a support office for first-generation college students to utilize if they have any questions or concerns. Please check out [First in Line](#), visit us in the Basement of Sid Richardson West Wing or email firstinline@baylor.edu. *[If you are a first gen college student, you may wish to add:] I was also the first in my family to graduate with a four-year bachelor's degree and can answer questions about experiences as a first-generation college student.*

Military Connected Students

Veterans, active duty military personnel, and dependents are encouraged to connect with the [VETS](#) program, a space dedicated to supporting our military-connected students. Please communicate, in advance if possible, any special circumstances (e.g., upcoming deployment, drill requirements, disability accommodations).

Health Resources for Students

Baylor University is strongly committed to addressing the mental health and wellness needs of students by providing access to on-campus healthcare resources.

Crises and Emergencies

BUCC Crisis Line: (254) 710-2467 (Business Hours/Non-Business Hours/Weekends)

Baylor Police Department: (254)710-222

MHMR Crisis Center: (254)867-6550

MHMR 24-Hour Emergency/Crisis Number: (254)752-3451

When home during academic breaks, when the counseling center is closed, please call your local resources. Some national numbers for support during a crisis:

National Hope Network Hotline: 1-800-SUICIDE (1-800-784-2433)

National Suicide Prevention Lifeline: 1-800-273-TALK (1-800-273-8255)

General Guidance for struggling students: Baylor University CARE Team Services

Located on the second floor of the Student Life Center, suite 207.

[The Department of CARE Team Services](#) is a team of case managers who work with students who are struggling with mental health issues, financial struggles, and anything else affecting a student's ability to be successful in the classroom.

Contact Information: (254) 710-2100 ; CareTeam@baylor.edu.

Students' Basic Needs (food & housing)

Food insecurity is defined as “a lack of consistent access to enough food for you to live an active, healthy life.” At Baylor, we want all students to have access to food resources that will support their holistic well-being and success. If you or someone you know experiences food insecurity at any time, you can find information on campus and community food resources by visiting [The StoreLinks to an external site.](#). You can also contact Store staff at 254-710-4931. For additional basic needs assistance, please reach out to [CASE](#) or the [Care Team](#).

Physical Health and Wellness, Psychiatric Evaluations, Pharmacy: Baylor Health Services Department

[Baylor Health ServicesLinks to an external site.](#) includes Primary Care, Psychiatry, Physical Therapy and Pharmacy and is staffed with fully certified and licensed physicians and nurse practitioners, as well as nurses and administrative staff. Appointments may be made by calling our main number or by logging into the health portal located on our website.

Contact Information: (254) 710-1010; Health_Services@baylor.edu.

Baylor University Counseling Center

[The Counseling CenterLinks to an external site.](#) seeks to foster wholeness for every student through caring relationships, cultural humility and integrated mental health services.

Located on the second floor of the Student Life Center.

For an appointment, go to our [websiteLinks to an external site.](#) to schedule an Initial Assessment or call (254) 710-2467.

Substance and Behavioral Addiction: Beauchamp Addiction & Recovery Center (BARC)

[The Beauchamp Addiction Recovery CenterLinks to an external site.](#) aims to support students in recovery from substance and behavioral addictions through an all-encompassing level of support approach that includes one-on-one mentorship, support groups, and social events open to all Baylor students.

Located in the East Village Residential Community (bottom floor of Teal Residential College).

Contact Information: (254)-710-7092; BARC@baylor.edu

Spiritual Life: Baylor University Office of Spiritual Life

[The Office of Spiritual LifeLinks to an external site.](#) offers programs, persons, and resources to nurture theological depth, spiritual wholeness, and missional living.

Located on the corner of 5th and Speight Street in the BoBo Spiritual Life Center.

Contact Information: (254) 710-3517; Spiritual_Life@baylor.edu

Copyright © 2006-2026 [Mengfei Ren](#), [Computer Science Department](#), [Baylor University](#), with some content taken from syllabi by Mr. Ernesto Quevedo Caballero, Dr. Paul Grabow and Dr. Eunjee Song.

Last major modification: August 13, 2026

Class Schedule

Below is the initial class schedule, check the updates in Canvas

Week	Date	Topics	Reading	Activities & Dues
Week 1	8/25/26	Course Intro	SE Ch.1	- Team Project Formation (09/03) - Tool Presentation Signup
	8/27/26	Software Engineering & Products	SE Products Ch.1	
Week 2	9/1/26	- Software Processes	SE Ch.2 & 3	- Design Studio 1: React
	9/3/26	- Agile SE	SE Products Ch.2	
Week 3	9/8/26	Requirement Engineering	SE Ch.4	- Design Studio 2: Case Modeling
	9/10/26		SE Products Ch.3	
Week 4	9/15/26	System Modeling	SE Ch.5	- Design Studio 3: Spring MVC
	9/17/26			
Week 5	9/22/26	Architecture Design + Design Patterns	SE Ch.6 & 7	- Design Studio 4: SSD
	9/24/26		SE Products Ch.4	
Week 6	9/29/26	Project Iteration 1 - Presentation	SE Ch.7	- Project Deliverable 1 (09/28)
	10/1/26	Design Patterns		
Week 7	10/6/26	No Class (Instructor in a conference)		- Design Studio 5: Design Patterns
	10/8/26			
Week 8	10/13/26	Testing	SE Ch.8	
	10/15/26	DevOps and Code Management	SE Products Ch.9 & 10	
Week 9	10/20/26	Configuration Management	SE Ch.10 & 25	
	10/22/26	Dependability		
Week 10	10/27/26	Reliability	SE Ch.11	- Project Deliverable 2 (10/28)
	10/29/26	Project Iteration 2 - Presentation	SE Products Ch.8	
Week 11	11/3/26	Software Safety & Security	SE Ch.12 & 13	
	11/5/26		SE Products Ch.7	
Week 12	11/10/26	Cloud-based Software & Microservices	SE Ch.17 & 18	
	11/12/26		SE Products Ch.5 & 6	
Week 13	11/17/26	System Engineering	SE Ch.19 & 20	
	11/19/26	Project Management		
Week 14	THANKSGIVING HOLIDAYS (11/23 - 11/27)			
Week 15	12/1/26	Final Exam Review		- Project Final Deliverable (12/11)
	12/3/26	Project Iteration 3 - Final Presentation		
Week 16	12/8/26			
	12/11/26	Final Exam (Cumulative) 2:00-3:15pm		- Baylor Course Evaluation